

Predictive N_T — Six-Component COSMO-SAC-2010

RESEARCH DIAGNOSTIC — NOT ACCEPTED — NOT RELEASE ELIGIBLE

Executive verdict

The calculation completed successfully for stage counts 1 through 10. No trial simultaneously satisfied all numerical, thermodynamic, component-balance, and saved product-target gates. Predictive N_T and established theoretical stages therefore remain unassigned.

Job status	COMPLETED — no runtime error
Trials completed	10 / 10
Diagnostic Predictive N_T	Not assigned
Established theoretical stages	Not established
Sulfur prediction	NOT CALCULABLE
Pilot validated	No
Calibration required	Yes
Release eligible	No

ENGINEERING INTERPRETATION

“Not accepted” is a scientific qualification verdict, not a software failure. The run provides diagnostic evidence showing where recovery, solvent carryover, PA, stability, closure, or saved product-target gates fail.

Completed 31/8/2026, 10:37:05 am IST

Frozen job: 1acd1a68-51a0-4877-94fc-0f199eb9f467

1. Frozen Stage 1 design basis

Saved authority used without live-input substitution

Project reference	111
RRBO grade	SN300
Feed rate	4000 L/h
Operating temperature	30 °C
Operating pressure	2.0
Phase configuration	nmp-continuous-rrbo-dispersed
Solvent / oil mass ratio	0.75
Maximum stage search	10
Feed sulfur	3500 ppm
Sulfur target	1000 ppm — NOT CALCULABLE

Feed composition — wt%

Basis	SAT	MONO	DI	POLY	PA	NMP
Feed	85	7	4	2	2	0

Saved product targets

Minimum NMP-free RRBO recovery	95%
Maximum NMP in total raffinate	0.5 wt%
Minimum saturates, NMP-free	90 wt%
Maximum total aromatics, NMP-free	5 wt%
Maximum PA, NMP-free	0.1 wt%

MODELED SOLVENT BASIS

Fresh solvent is pure modeled NMP at the saved S/O ratio. Commercial purity and water specifications are audit-only and do not add components.

2. Six-component trial comparison

All six modeled families shown for every trial

Stages	SAT extraction %	MONO extraction %	DI extraction %	POLY extraction %	PA extraction %	NMP extraction %
1	NOT CALC.	47.998	49.118	52.026	66.736	NOT CALC.
2	NOT CALC.	65.143	66.288	70.142	86.429	NOT CALC.
3	NOT CALC.	74.08	75.241	79.456	94.134	NOT CALC.
4	NOT CALC.	79.468	80.691	85.025	97.404	NOT CALC.
5	NOT CALC.	83.027	84.334	88.669	98.84	NOT CALC.
6	NOT CALC.	85.533	86.93	91.202	99.479	NOT CALC.
7	NOT CALC.	87.381	88.867	93.039	99.766	NOT CALC.
8	NOT CALC.	88.43	89.587	93.111	99.766	NOT CALC.
9	NOT CALC.	86.567	83.709	75.279	80.21	NOT CALC.
10	NOT CALC.	87.257	83.985	75.803	80.7	NOT CALC.

Final raffinate composition — mole fraction

Stages	SAT	MONO	DI	POLY	PA	NMP
1	0.79265	0.05086	0.02404	0.00797	0.00275	0.12173
2	0.82539	0.03577	0.01671	0.0052	0.00118	0.11575
3	0.84322	0.02729	0.01259	0.00367	0.00052	0.11271
4	0.85421	0.02195	0.00997	0.00272	0.00024	0.11091
5	0.86157	0.01834	0.00817	0.00208	0.00011	0.10973
6	0.8668	0.01574	0.00687	0.00163	4.80e-5	0.10891
7	0.87069	0.0138	0.00588	0.00129	2.17e-5	0.10831
8	0.87244	0.01269	0.00552	0.00128	2.17e-5	0.10805
9	0.86077	0.01451	0.0085	0.00453	0.00181	0.10989
10	0.86187	0.01381	0.00838	0.00445	0.00177	0.10971

3. Trial 1 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 2.012e-16

Coupled solver CONVERGED · Maximum scaled equation residual 2.653e-14 · Multistart outlet difference 4.394e-11

Blockers: POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	87.6898
maximum Nmp Raffinate Wt	FAIL	0.5	7.5728
minimum Raffinate Saturates Wt	PASS	90	91.675
maximum Raffinate Total Aromatics Wt	FAIL	5	7.5663
maximum Raffinate Polar Aromatics Wt	FAIL	0.1	0.7587
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	47.9982	49.1181	52.0264	66.7365	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.7195e-1	3.0285e-2	1.4313e-2	4.7440e-3	1.6403e-3	7.2476e-2
final Extract	2.7066e-2	2.7954e-2	1.3817e-2	5.1447e-3	3.2909e-3	6.8410e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	2.012e-16	1.041e-17	3.469e-18	0.000e+0	8.674e-19	-1.110e-16

3. Trial 1 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	2.012e-16	FAIL	2.653e-14	2.687e-3	4.210e-3	-4.312e-1	-4.312e-1

3. Trial 1 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.792649	0.050865	0.024039	0.007968	0.002755	0.121725
1 E	0.03555	0.036715	0.018147	0.006757	0.004322	0.898508

3. Trial 2 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 7.633e-17

Coupled solver CONVERGED · Maximum scaled equation residual 6.384e-14 · Multistart outlet difference 5.453e-4

Blockers: MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	84.4392
maximum Nmp Raffinate Wt	FAIL	0.5	7.1591
minimum Raffinate Saturates Wt	PASS	90	94.4847
maximum Raffinate Total Aromatics Wt	FAIL	5	5.1938
maximum Raffinate Polar Aromatics Wt	FAIL	0.1	0.3214
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	65.1432	66.2883	70.1417	86.429	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6838e-1	2.0300e-2	9.4831e-3	2.9526e-3	6.6921e-4	6.5683e-2
final Extract	3.0633e-2	3.7939e-2	1.8647e-2	6.9361e-3	4.2620e-3	6.9089e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	7.633e-17	6.939e-18	3.469e-18	4.337e-18	1.735e-18	0.000e+0

3. Trial 2 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	1.110e-16	FAIL	1.577e-14	3.334e-3	5.111e-3	-5.340e-1	-5.340e-1
2	2.776e-17	FAIL	6.384e-14	1.167e-3	1.847e-3	-2.112e-1	-2.112e-1

3. Trial 2 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.765957	0.062719	0.030669	0.010183	0.003437	0.127034
1 E	0.03881	0.048066	0.023624	0.008788	0.0054	0.875313
2 R	0.825387	0.035773	0.016711	0.005203	0.001179	0.115747
2 E	0.032155	0.024235	0.012382	0.004346	0.001859	0.925023

3. Trial 3 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 4.215e-16

Coupled solver CONVERGED · Maximum scaled equation residual 2.776e-14 · Multistart outlet difference 6.758e-4

Blockers: MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	82.7866
maximum Nmp Raffinate Wt	FAIL	0.5	6.9471
minimum Raffinate Saturates Wt	PASS	90	95.9741
maximum Raffinate Total Aromatics Wt	PASS	5	3.8842
maximum Raffinate Polar Aromatics Wt	FAIL	0.1	0.1417
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	74.08	75.2414	79.4562	94.1343	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6646e-1	1.5096e-2	6.9646e-3	2.0315e-3	2.8925e-4	6.2349e-2
final Extract	3.2561e-2	4.3144e-2	2.1165e-2	7.8572e-3	4.6420e-3	6.9423e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	5.551e-17	6.939e-18	-6.939e-18	-1.735e-18	-4.215e-16	0.000e+0

3. Trial 3 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	1.110e-16	FAIL	1.943e-14	3.549e-3	5.342e-3	-5.648e-1	-5.648e-1
2	1.184e-16	FAIL	2.776e-14	1.550e-3	2.438e-3	-4.050e-1	-4.050e-1
3	2.637e-16	FAIL	1.465e-14	5.203e-4	8.257e-4	-1.254e-2	-1.254e-2

3. Trial 3 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.753957	0.06782	0.033795	0.011213	0.003667	0.12955
1 E	0.040519	0.053688	0.026338	0.009778	0.005776	0.8639
2 R	0.798354	0.048413	0.02363	0.007357	0.001571	0.120675
2 E	0.034753	0.034297	0.017737	0.006209	0.002466	0.904537
3 R	0.843218	0.027288	0.01259	0.003672	0.000523	0.112708
3 E	0.030574	0.017959	0.009259	0.00305	0.000828	0.938331

3. Trial 4 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 1.388e-17

Coupled solver CONVERGED · Maximum scaled equation residual 4.402e-14 · Multistart outlet difference 5.045e-4

Blockers: MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	81.8134
maximum Nmp Raffinate Wt	FAIL	0.5	6.8204
minimum Raffinate Saturates Wt	PASS	90	96.8697
maximum Raffinate Total Aromatics Wt	PASS	5	3.0669
maximum Raffinate Polar Aromatics Wt	PASS	0.1	0.0635
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	79.4683	80.6906	85.0249	97.4042	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6527e-1	1.1957e-2	5.4317e-3	1.4808e-3	1.2801e-4	6.0409e-2
final Extract	3.3743e-2	4.6282e-2	2.2698e-2	8.4079e-3	4.8032e-3	6.9616e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	1.388e-17	-1.388e-17	0.000e+0	-3.469e-18	1.735e-18	0.000e+0

3. Trial 4 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	1.110e-16	FAIL	1.155e-14	3.623e-3	5.376e-3	-5.784e-1	-5.784e-1
2	1.110e-16	FAIL	1.599e-14	1.694e-3	2.655e-3	-4.723e-1	-4.723e-1
3	9.368e-17	FAIL	1.066e-14	7.151e-4	1.128e-3	-2.764e-1	-2.764e-1
4	1.041e-17	PASS	4.402e-14	2.341e-4	3.727e-4	3.505e-18	-1.143e-19

3. Trial 4 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.747321	0.070567	0.035597	0.011794	0.003745	0.130976
1 E	0.04155	0.05699	0.02795	0.010353	0.005915	0.857242
2 R	0.783962	0.054843	0.027483	0.008537	0.001718	0.123458
2 E	0.0363	0.039886	0.020804	0.007256	0.002694	0.89306
3 R	0.818539	0.039257	0.018982	0.005527	0.000719	0.116976
3 E	0.032692	0.026788	0.014091	0.004622	0.001133	0.920675
4 R	0.854213	0.021953	0.009972	0.002719	0.000235	0.110908
4 E	0.029678	0.014215	0.007306	0.002251	0.000373	0.946176

3. Trial 5 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 2.613e-13

Coupled solver CONVERGED · Maximum scaled equation residual 1.058e-13 · Multistart outlet difference 5.282e-4

Blockers: MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	81.1828
maximum Nmp Raffinate Wt	FAIL	0.5	6.7371
minimum Raffinate Saturates Wt	PASS	90	97.4569
maximum Raffinate Total Aromatics Wt	PASS	5	2.5145
maximum Raffinate Polar Aromatics Wt	PASS	0.1	0.0286
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	83.0273	84.3336	88.6689	98.8397	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6449e-1	9.8848e-3	4.4070e-3	1.1205e-3	5.7218e-5	5.9158e-2
final Extract	3.4531e-2	4.8354e-2	2.3723e-2	8.7682e-3	4.8740e-3	6.9742e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	-3.886e-16	-6.939e-18	-3.469e-16	-1.613e-16	-2.613e-13	5.551e-16

3. Trial 5 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	4.272e-15	FAIL	7.372e-14	3.644e-3	5.350e-3	-5.859e-1	-5.859e-1
2	1.269e-14	FAIL	2.254e-14	1.746e-3	2.732e-3	-5.047e-1	-5.047e-1
3	3.101e-14	FAIL	9.326e-15	7.930e-4	1.248e-3	-3.773e-1	-3.773e-1
4	6.980e-14	FAIL	5.285e-14	3.281e-4	5.186e-4	-1.564e-1	-1.564e-1
5	1.435e-13	PASS	2.809e-14	1.058e-4	1.686e-4	2.586e-18	-3.534e-18

3. Trial 5 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.743173	0.072255	0.036765	0.012159	0.003767	0.131882
1 E	0.042231	0.059137	0.029013	0.010723	0.005961	0.852935
2 R	0.775168	0.058658	0.029922	0.009266	0.001772	0.125214
2 E	0.037311	0.043396	0.022783	0.007915	0.002779	0.885816
3 R	0.803893	0.046054	0.022943	0.006654	0.000798	0.119657
3 E	0.034072	0.032154	0.017146	0.005595	0.001254	0.909779
4 R	0.832133	0.03297	0.015708	0.004267	0.000329	0.114593
4 E	0.03145	0.02201	0.01159	0.003552	0.000519	0.930878
5 R	0.861574	0.018335	0.008174	0.002078	0.000106	0.109732
5 E	0.02911	0.011752	0.005975	0.001718	0.000169	0.951275

3. Trial 6 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 1.485e-15

Coupled solver CONVERGED · Maximum scaled equation residual 8.615e-14 · Multistart outlet difference 4.037e-4

Blockers: MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	80.7451
maximum Nmp Raffinate Wt	FAIL	0.5	6.6785
minimum Raffinate Saturates Wt	PASS	90	97.8675
maximum Raffinate Total Aromatics Wt	PASS	5	2.1196
maximum Raffinate Polar Aromatics Wt	PASS	0.1	0.0129
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	85.5326	86.9296	91.2017	99.4792	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6393e-1	8.4257e-3	3.6767e-3	8.7004e-4	2.5681e-5	5.8292e-2
final Extract	3.5088e-2	4.9813e-2	2.4453e-2	9.0187e-3	4.9055e-3	6.9828e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	0.000e+0	2.082e-17	0.000e+0	0.000e+0	-1.485e-15	0.000e+0

3. Trial 6 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	1.301e-17	FAIL	3.786e-14	3.645e-3	5.309e-3	-5.904e-1	-5.904e-1
2	1.110e-16	FAIL	1.554e-14	1.763e-3	2.754e-3	-5.232e-1	-5.232e-1
3	8.674e-17	FAIL	1.377e-14	8.234e-4	1.293e-3	-4.287e-1	-4.287e-1
4	1.869e-16	FAIL	2.232e-14	3.673e-4	5.788e-4	-2.859e-1	-2.859e-1
5	4.042e-16	FAIL	8.615e-14	1.499e-4	2.374e-4	-4.696e-2	-4.696e-2
6	7.666e-16	PASS	3.275e-14	4.782e-5	7.642e-5	-2.549e-18	-3.229e-18

3. Trial 6 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.740361	0.073384	0.037581	0.012406	0.003767	0.132501
1 E	0.042709	0.060633	0.029764	0.010977	0.005971	0.849945
2 R	0.769294	0.061154	0.0316	0.009752	0.001789	0.126412
2 E	0.038017	0.045785	0.024161	0.00836	0.002805	0.88087
3 R	0.794299	0.050373	0.025626	0.007398	0.000829	0.121475
3 E	0.035034	0.035727	0.019245	0.006246	0.001301	0.902447
4 R	0.81801	0.039725	0.019538	0.005278	0.000368	0.11708
4 E	0.032682	0.027068	0.014496	0.004413	0.00058	0.920762
5 R	0.841831	0.028423	0.01329	0.003361	0.00015	0.112946
5 E	0.030627	0.018705	0.00977	0.00279	0.000238	0.937871
6 R	0.866803	0.015743	0.00687	0.001626	4.80e-5	0.108912
6 E	0.028722	0.01002	0.005014	0.001343	7.64e-5	0.954825

3. Trial 7 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 7.577e-13

Coupled solver CONVERGED · Maximum scaled equation residual 2.856e-13 · Multistart outlet difference 4.083e-4

Blockers: MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_LOCAL_STABILITY_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	80.4253
maximum Nmp Raffinate Wt	FAIL	0.5	6.6354
minimum Raffinate Saturates Wt	PASS	90	98.1691
maximum Raffinate Total Aromatics Wt	PASS	5	1.8251
maximum Raffinate Polar Aromatics Wt	PASS	0.1	0.0058
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	87.3814	88.8673	93.0393	99.7659	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6352e-1	7.3489e-3	3.1316e-3	6.8833e-4	1.1543e-5	5.7659e-2
final Extract	3.5502e-2	5.0890e-2	2.4998e-2	9.2004e-3	4.9197e-3	6.9891e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	-4.580e-16	-9.714e-17	-4.892e-16	-2.429e-16	-7.577e-13	5.551e-16

3. Trial 7 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	2.495e-15	FAIL	7.994e-15	3.639e-3	5.267e-3	-5.935e-1	-5.935e-1
2	7.453e-15	FAIL	1.155e-14	1.765e-3	2.754e-3	-5.350e-1	-5.350e-1
3	1.833e-14	FAIL	1.055e-14	8.338e-4	1.308e-3	-4.591e-1	-4.591e-1
4	4.206e-14	FAIL	2.620e-14	3.832e-4	6.029e-4	-3.545e-1	-3.545e-1
5	9.366e-14	FAIL	2.043e-14	1.687e-4	2.663e-4	-2.002e-1	-2.002e-1
6	2.063e-13	PASS	3.475e-14	6.824e-5	1.081e-4	3.121e-18	-2.555e-18
7	3.874e-13	FAIL	1.172e-13	2.168e-5	3.457e-5	4.151e-18	-1.119e-18

3. Trial 7 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.738343	0.074186	0.038181	0.01258	0.00376	0.132949
1 E	0.043063	0.061728	0.030322	0.01116	0.005967	0.84776
2 R	0.76512	0.062899	0.032821	0.010094	0.001791	0.127274
2 E	0.038536	0.047506	0.025174	0.008677	0.002809	0.877298
3 R	0.787574	0.053333	0.027558	0.007916	0.00084	0.122779
3 E	0.035739	0.038261	0.020772	0.006704	0.001316	0.897207
4 R	0.808275	0.044242	0.022266	0.005978	0.000384	0.118855
4 E	0.03358	0.030591	0.01659	0.005015	0.000604	0.91362
5 R	0.828494	0.034968	0.016899	0.004243	0.000169	0.115227
5 E	0.03173	0.023443	0.012481	0.003535	0.000267	0.928545
6 R	0.849061	0.025001	0.011438	0.002687	6.83e-5	0.111745
6 E	0.030042	0.016289	0.008388	0.002227	0.000108	0.942946
7 R	0.870689	0.013805	0.005883	0.001293	2.17e-5	0.10831
7 E	0.02844	0.008743	0.004289	0.001067	3.46e-5	0.957426

3. Trial 8 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 4.687e-4

Coupled solver NOT CONVERGED · Maximum scaled equation residual 8.999e-5 · Multistart outlet difference 6.572e-5

Blockers: COUPLED_SOLVER_CLOSURE_FAILED · MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_LOCAL_STABILITY_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	80.2476
maximum Nmp Raffinate Wt	FAIL	0.5	6.6158
minimum Raffinate Saturates Wt	PASS	90	98.2942
maximum Raffinate Total Aromatics Wt	PASS	5	1.7
maximum Raffinate Polar Aromatics Wt	PASS	0.1	0.0058
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	88.4303	89.5866	93.1109	99.7662	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6308e-1	6.7381e-3	2.9293e-3	6.8124e-4	1.1531e-5	5.7349e-2
final Extract	3.5941e-2	5.1970e-2	2.5490e-2	9.3895e-3	5.1230e-3	6.9920e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	-3.666e-6	-4.687e-4	-2.892e-4	-1.820e-4	-2.033e-4	1.979e-5

3. Trial 8 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	1.914e-5	FAIL	9.728e-7	3.767e-3	5.360e-3	-5.953e-1	-5.953e-1
2	2.491e-5	FAIL	9.623e-7	1.975e-3	3.076e-3	-5.432e-1	-5.432e-1
3	1.092e-4	FAIL	1.156e-6	1.088e-3	1.706e-3	-4.803e-1	-4.803e-1
4	8.636e-5	FAIL	3.586e-7	5.818e-4	9.146e-4	-4.005e-1	-4.005e-1
5	7.966e-5	FAIL	6.248e-7	2.928e-4	4.612e-4	-2.916e-1	-2.916e-1
6	7.566e-5	FAIL	8.199e-7	1.402e-4	2.221e-4	-1.326e-1	-1.326e-1
7	1.221e-4	FAIL	3.567e-7	6.088e-5	9.794e-5	-5.986e-18	1.042e-18
8	1.065e-4	FAIL	5.428e-7	2.156e-5	3.458e-5	1.753e-16	-4.102e-18

3. Trial 8 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.736414	0.074876	0.038681	0.012759	0.003897	0.133373
1 E	0.043453	0.062832	0.030818	0.011352	0.006194	0.845351
2 R	0.761223	0.064384	0.033849	0.010455	0.002008	0.128081
2 E	0.039082	0.049124	0.026057	0.009016	0.003152	0.87357
3 R	0.781407	0.055808	0.029215	0.008484	0.001098	0.123989
3 E	0.036457	0.040542	0.022115	0.00721	0.001721	0.891955
4 R	0.7996	0.047934	0.024647	0.006762	0.000585	0.120473
4 E	0.034454	0.033631	0.018448	0.005693	0.000918	0.906856
5 R	0.817058	0.040098	0.020075	0.005221	0.000294	0.117255
5 E	0.032754	0.027339	0.014898	0.004366	0.000462	0.920181
6 R	0.834581	0.031878	0.015411	0.003813	0.000141	0.114177
6 E	0.031216	0.021182	0.011357	0.003171	0.000222	0.932852
7 R	0.852769	0.022943	0.010577	0.002507	6.17e-5	0.111143
7 E	0.02976	0.014871	0.007749	0.002076	9.80e-5	0.945447
8 R	0.872436	0.012694	0.005519	0.001283	2.17e-5	0.108045
8 E	0.028321	0.008021	0.004023	0.001059	3.47e-5	0.958541

3. Trial 9 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 3.909e-3

Coupled solver NOT CONVERGED · Maximum scaled equation residual 1.315e-3 · Multistart outlet difference 1.055e-3

Blockers: COUPLED_SOLVER_CLOSURE_FAILED · MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	81.549
maximum Nmp Raffinate Wt	FAIL	0.5	6.7194
minimum Raffinate Saturates Wt	PASS	90	96.9562
maximum Raffinate Total Aromatics Wt	PASS	5	2.5584
maximum Raffinate Polar Aromatics Wt	FAIL	0.1	0.4854
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	86.5671	83.7092	75.2786	80.21	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6418e-1	7.8232e-3	4.5826e-3	2.4446e-3	9.7588e-4	5.9258e-2
final Extract	3.5883e-2	5.4325e-2	2.4668e-2	8.0434e-3	4.9081e-3	6.9700e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	-1.051e-3	-3.909e-3	-1.121e-3	-5.994e-4	-9.528e-4	3.180e-4

3. Trial 9 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	5.967e-4	FAIL	5.024e-6	3.604e-3	5.068e-3	-6.260e-1	-6.260e-1
2	9.710e-4	FAIL	5.784e-6	2.134e-3	3.319e-3	-6.025e-1	-6.025e-1
3	9.718e-4	FAIL	9.166e-6	2.054e-3	3.210e-3	-5.630e-1	-5.630e-1
4	7.028e-4	FAIL	9.602e-6	2.276e-3	3.564e-3	-5.040e-1	-5.040e-1
5	5.096e-4	FAIL	5.495e-6	2.407e-3	3.782e-3	-4.225e-1	-4.225e-1
6	6.163e-4	FAIL	1.000e-5	2.485e-3	3.919e-3	-3.138e-1	-3.138e-1
7	9.234e-4	FAIL	1.179e-5	2.596e-3	4.114e-3	-1.624e-1	-1.624e-1
8	1.353e-3	FAIL	1.085e-5	2.526e-3	4.030e-3	-1.909e-18	8.095e-19
9	1.784e-3	FAIL	1.317e-5	1.779e-3	2.855e-3	4.932e-18	-9.781e-19

3. Trial 9 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.737443	0.077396	0.037387	0.010913	0.003723	0.133138
1 E	0.043504	0.065862	0.029907	0.009752	0.00595	0.845025
2 R	0.761123	0.069046	0.031829	0.007755	0.002173	0.128072
2 E	0.039254	0.053658	0.024601	0.006716	0.003426	0.872345
3 R	0.777449	0.062246	0.027249	0.006252	0.002091	0.124713
3 E	0.037143	0.046541	0.020768	0.005347	0.003287	0.886915
4 R	0.789288	0.05585	0.023921	0.006252	0.002323	0.122366
4 E	0.035832	0.040741	0.018088	0.00531	0.003648	0.896381
5 R	0.798905	0.04931	0.021739	0.007056	0.002461	0.12053
5 E	0.034848	0.035227	0.016346	0.005964	0.003864	0.903751
6 R	0.808364	0.042349	0.020039	0.007922	0.002542	0.118784
6 E	0.033945	0.029661	0.014998	0.006671	0.003995	0.910729
7 R	0.82004	0.034566	0.017877	0.008164	0.002661	0.116693
7 E	0.032902	0.023713	0.013319	0.00685	0.004189	0.919028
8 R	0.836564	0.025517	0.014296	0.007193	0.00259	0.11384
8 E	0.031477	0.017064	0.010588	0.006008	0.004092	0.930772
9 R	0.860767	0.014507	0.008498	0.004533	0.00181	0.109886
9 E	0.029457	0.009362	0.006239	0.003762	0.002878	0.948302

3. Trial 10 — complete diagnostic overview

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates FAIL · maximum component-balance residual 4.136e-3

Coupled solver NOT CONVERGED · Maximum scaled equation residual 1.264e-3 · Multistart outlet difference 1.839e-3

Blockers: COUPLED_SOLVER_CLOSURE_FAILED · MULTISTART_BRANCH_REPRODUCTION_FAILED · MULTISTART_SECONDARY_CLOSURE_FAILED · POST_SPLIT_TPD_STABILITY_FAILED

Product metrics and target checks

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	FAIL	95	81.3026
maximum Nmp Raffinate Wt	FAIL	0.5	6.7073
minimum Raffinate Saturates Wt	PASS	90	97.045
maximum Raffinate Total Aromatics Wt	PASS	5	2.4803
maximum Raffinate Polar Aromatics Wt	FAIL	0.1	0.4748
target Raffinate Sulfur Ppm	NOT CALCULABLE	1000	—

Six-component extraction diagnostics — %

Basis	SAT	MONO	DI	POLY	PA	NMP
Extraction	NOT CALC.	87.2571	83.9852	75.8034	80.7004	NOT CALC.

Stream	SAT	MONO	DI	POLY	PA	NMP
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	7.5657e-1
final Raffinate	4.6321e-1	7.4214e-3	4.5050e-3	2.3927e-3	9.5171e-4	5.8965e-2
final Extract	3.5981e-2	5.4954e-2	2.4264e-2	8.1148e-3	5.0064e-3	6.9751e-1

Overall six-component balance residual — mol

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	-1.699e-4	-4.136e-3	-6.391e-4	-6.188e-4	-1.027e-3	9.587e-5

3. Trial 10 — internal stage diagnostics

Local balance, isoactivity, and independent stability/TPD diagnostics for every internal stage

Stage	Local balance	Local gate	Isoactivity	R min eig	E min eig	R TPD min	ETPD min
1	5.359e-4	FAIL	5.628e-6	3.664e-3	5.095e-3	-6.312e-1	-6.312e-1
2	8.742e-4	FAIL	3.187e-6	2.184e-3	3.394e-3	-6.131e-1	-6.131e-1
3	1.025e-3	FAIL	4.373e-6	2.062e-3	3.222e-3	-5.807e-1	-5.806e-1
4	9.626e-4	FAIL	8.412e-6	2.243e-3	3.512e-3	-5.319e-1	-5.319e-1
5	5.321e-4	FAIL	5.663e-6	2.360e-3	3.704e-3	-4.671e-1	-4.671e-1
6	5.992e-4	FAIL	5.312e-6	2.396e-3	3.771e-3	-3.833e-1	-3.833e-1
7	6.482e-4	FAIL	5.104e-6	2.481e-3	3.920e-3	-2.718e-1	-2.718e-1
8	1.102e-3	FAIL	9.288e-6	2.613e-3	4.147e-3	-1.181e-1	-1.181e-1
9	1.362e-3	FAIL	1.748e-5	2.525e-3	4.032e-3	-8.235e-18	1.423e-18
10	1.716e-3	FAIL	1.119e-5	1.741e-3	2.796e-3	8.182e-19	-2.232e-18

3. Trial 10 — six-component stage outlets

Raffinate and extract leaving-phase mole fractions for every internal stage

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.737645	0.077826	0.036674	0.01098	0.003787	0.133088
1 E	0.043569	0.066543	0.029381	0.009826	0.006062	0.844617
2 R	0.761384	0.069862	0.030618	0.0079	0.002225	0.128011
2 E	0.039284	0.054532	0.02369	0.006848	0.003511	0.872135
3 R	0.777557	0.063517	0.025802	0.006342	0.002099	0.124683
3 E	0.037171	0.047713	0.019682	0.005428	0.003303	0.886704
4 R	0.788773	0.057759	0.022584	0.006141	0.002289	0.122455
4 E	0.035906	0.04237	0.017092	0.00522	0.003597	0.895815
5 R	0.796907	0.052196	0.020904	0.006688	0.002411	0.120894
5 E	0.035052	0.03758	0.015739	0.00566	0.003787	0.902182
6 R	0.803969	0.046406	0.020107	0.007486	0.002449	0.119582
6 E	0.034349	0.032842	0.015079	0.006314	0.003846	0.907569
7 R	0.811895	0.039973	0.019319	0.008128	0.002539	0.118145
7 E	0.033619	0.02781	0.014438	0.006836	0.003992	0.913306
8 R	0.822672	0.032673	0.017555	0.008186	0.002679	0.116234
8 E	0.032677	0.022309	0.013067	0.006864	0.00422	0.920863
9 R	0.838487	0.024211	0.014079	0.007114	0.002588	0.113521
9 E	0.031323	0.016139	0.010421	0.00594	0.004092	0.932085
10 R	0.861871	0.013809	0.008382	0.004452	0.001771	0.109715
10 E	0.029372	0.008895	0.006152	0.003694	0.002818	0.949069

5. Numerical diagnostics

Scientific solver completion is distinct from trial qualification

Stages	Max scaled residual	Mole balance	Mass balance	Multistart difference	Gate
1	2.65e-14	2.01e-16	3.43e-14	4.39e-11	FAIL
2	6.38e-14	7.63e-17	1.30e-14	0	FAIL
3	2.78e-14	4.22e-16	1.71e-13	0	FAIL
4	4.40e-14	1.39e-17	2.36e-15	0	FAIL
5	1.06e-13	2.61e-13	1.06e-10	0	FAIL
6	8.62e-14	1.48e-15	6.02e-13	0	FAIL
7	2.86e-13	7.58e-13	3.07e-10	0	FAIL
8	9.00e-5	0	0.08	6.57e-5	FAIL
9	0	0	0.47	0	FAIL
10	0	0	0.5	0	FAIL

Qualification blockers

- 1: POST_SPLIT_TPD_STABILITY_FAILED
- 2: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 3: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 4: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 5: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 6: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 7: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 8: COUPLED_SOLVER_CLOSURE_FAILED; MULTISTART_BRANCH_REPRODUCTION_FAILED
- 9: COUPLED_SOLVER_CLOSURE_FAILED; MULTISTART_BRANCH_REPRODUCTION_FAILED
- 10: COUPLED_SOLVER_CLOSURE_FAILED; MULTISTART_BRANCH_REPRODUCTION_FAILED

6. Governance and provenance

Immutable identifiers for the frozen completed snapshot

CONTROLLED RESEARCH OUTPUT — CALIBRATION REQUIRED

Thermodynamic classification	SIX_COMPONENT_COSMO_SAC_2010_RESEARCH_DIAGNOSTIC
Model SHA-256	9666ff67194102ca9c3dc139695bead8ff133fbfe24c2d786ce1deca49456196
Engine SHA-256	5d9ab380d8dcadaaed3a8ada30dec216700f8215f8649549491c456a9672057d
Input SHA-256	93486a655df80b5d6c61d2e3400fbe8b9610697ab50edab0bf893424199197df
Stage 1 authority SHA-256	—
Six-component basis SHA-256	7683442c0419659d83320cfa4433fe8cf298f285f596c4676cf39bb5042838fc
Six-component binding SHA-256	d71161802f9294719416b209739f7918c27cda87a716162fe9308d948271049a
Job ID	1acd1a68-51a0-4877-94fc-0f199eb9f467

Mandatory limitations

- Sulfur removal is NOT CALCULABLE from this aromatic-transfer model.
- The PA representative is non-sulfur-bearing and is not a sulfur surrogate.
- No trial is pilot validated, accepted for design, or release eligible.
- No theoretical-stage count is established by this run.
- Calibration and direct matching NMP/RRBO evidence remain required.

DOCUMENT CONTROL

Generated only from the persisted completed job snapshot. No live design inputs were read and no scientific calculation was rerun for this report.